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Interaction of cloud-like droplets with the vortical structure

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ABSTRACT

Over the last decade, considerable progress has been made in understanding cloud formation and rain initiation. However, the development of cloud droplets into raindrops, which initiate precipitation in warm clouds, is still poorly understood. The fast development of these cloud droplets into raindrops is thought to be caused by collision and coalescence, which is significantly influenced by background turbulent flows. Thus, to better understand these collisions and coalescence, in the present study, we experimentally investigate the interaction of cloud-like droplets with the vortical structures that model eddies present in turbulent flows. We utilise a flow diagnostic technique, time-resolved particle image velocimetry (PIV), particle tracking velocimetry (PTV), and high-speed shadowgraphy, to study the interaction. The findings show that droplets are preferentially concentrated in the regions of low vorticity migrating from the high-vorticity regions, due to the vorticity field. voronoï analysis is used to quantify this preferential concentration, and we find that vortices lead to cluster formation. This clustering eventually may enhance the likelihood of collisions and subsequent coalescence through the formation of caustics.

Keywords: Cloud microphysics, vortices, droplets.

I. INTRODUCTION

The formation and characteristics of atmospheric clouds are highly complex and essential to our planet's climate system. Clouds can be made up of water droplets and/or ice crystals floating in the atmosphere. These clouds are essential to the hydrological cycle and preserve the Earth's radiation budget, absorbing long-wave radiation and reflecting short-wave solar radiation. Therefore, a thorough understanding of the microphysics of cloud formation is essential for accurately predicting future climate systems.

A schematic representation of cloud formation is shown in Figure 1. Solar radiation heats the surface of the liquid water, causing it to evaporate. Because of the density difference, this water vapor and the aerosol particles (an air parcel) ascend adiabatically in the atmosphere. The air parcel cools and expands as it ascends because of the decreased pressure. The moment this air parcel reaches the lifting condensation level (LCL) of the atmospheric boundary layer (ABL), it becomes saturated, and water vapor condenses. Thus, the cloud condensation nuclei are first created by nucleation ($\sim 1\,\mu$m), and then they increase in size as a result of further condensation of water vapor ($\sim 10\,\mu$m) [1]. Once these droplets are large enough ($\sim 100\,\mu$m), they start to fall under the effect of gravity. Subsequently, a raindrop is formed (~ 1 mm), from this falling the cloud droplet ($\sim 100\,\mu$m)

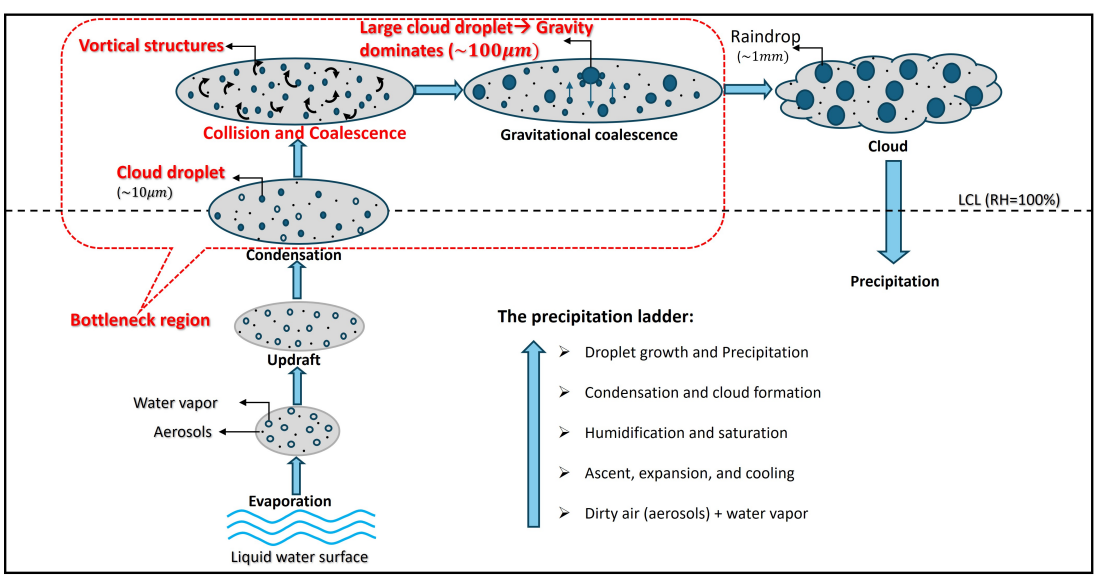


Figure 1: Schematic of cloud formation and rain initiation

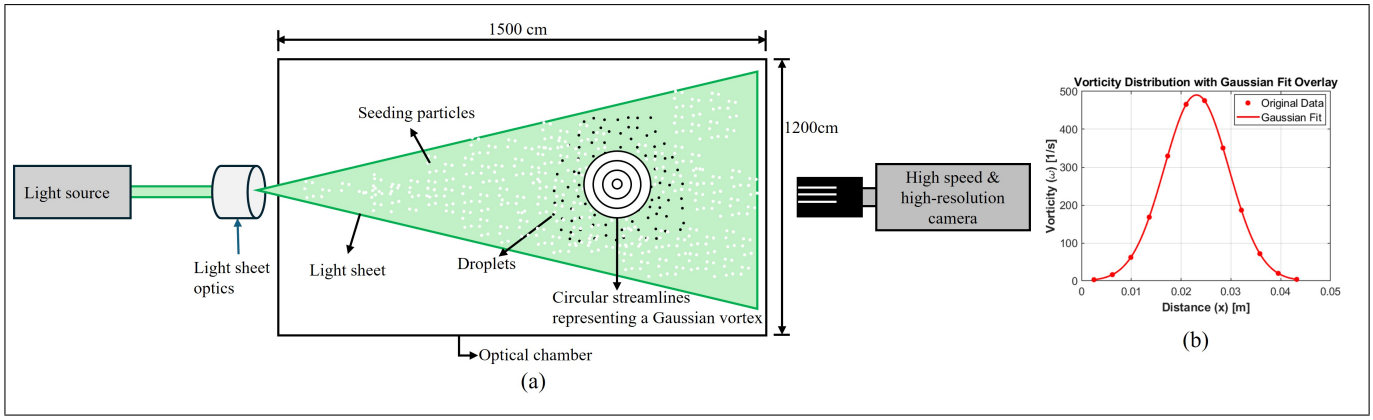


Figure 2: **(a) Schematic diagram of the experimental facility; (b) vorticity distribution along the vortex.**

because of gravitational collision and coalescence. However, the growth of cloud droplets from about 10μm to 100μm, known as the "bottleneck problem" in the atmospheric cloud formation, is poorly understood and is the focus of the present study. Atmospheric turbulence is hypothesized to facilitate growth and reduce the time required for raindrop formation [2], [3].

The atmospheric turbulence helps to grow condensation drops and shortens the time for raindrop formation. Turbulence can cause cloud droplets to collide and alter their electric charge; droplets with opposing charges will attract, merge, and increase in size until they descend under the influence of gravity. The collision rate within atmospheric clouds is influenced by the distribution of droplet sizes and the properties of the vortical structures in turbulent flow. Therefore, to fully understand the cloud formation, it is important to first study the collision and coalescence of the cloud droplets due to the presence of turbulence in the ABL. Various numerical studies have been conducted to understand the interaction of cloud droplets with turbulent flows [4], [5]. Nonetheless, numerous challenges accompany these techniques. For instance, the DNS study of droplet growth in turbulence within parameter ranges related to atmospheric clouds necessitates substantial computational resources due to the exceedingly high Reynolds number. The challenges can be addressed through an experimental study of this problem, facilitated by improved control of relevant parameters at a higher equivalent Reynolds number. Several experimental studies on droplet growth have been conducted within parameter ranges pertinent to atmospheric clouds. Thus, the present research focuses on experimentally studying the interaction of vortical structures with droplets to better understand the dominant growth of the droplet as a result of turbulence after a specific threshold diameter. The growth would be the function of the characteristics of the vortical structures and droplets present in the ABL, which comprises size, strength & the life cycle of the vortical structures, and initial droplet size distribution & their number density. Flow diagnostic techniques, such as shadowgraphy, multiphase time-resolved PIV, and PTV, will be utilized to study the complexity of clustering and collision in clouds.

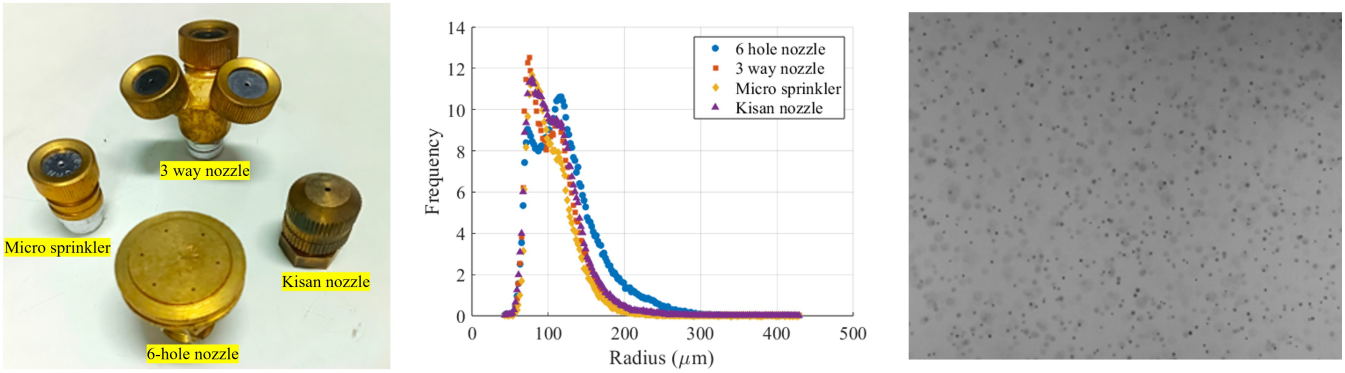
II. EXPERIMENTAL METHODOLOGY

An idealistic description of clouds could be the turbulent areas of fluids, namely water vapor and air. These regions are full of eddies/vortices ranging from a few millimeters to hundreds of meters. Such eddies, via the process of collision and coalescence, aid in the formation of bigger water droplets. Thus, the droplet dynamics in a cloud may be analyzed as an interaction between these vortices and droplets, as shown in the figure2, where we simplified the complex problem by making a vortex and seeing how it interacts with droplets, since it is challenging to study actual clouds in order to comprehend the physics behind them. We utilize the time-resolved particle image velocimetry (TR-PIV) to characterize this vortex, seeded by the fog particles with a mean size of 1 micron. The experiments were performed using the Photron UX-Mini 100, a high-speed and high-resolution camera, at a rate of 4000 Hz. The circular streamlines shown in figure2 represent an idealized vortical structure, which follows the Gaussian vorticity distribution within the vortex core.

We pass the pressurized water through the spray nozzles to generate multiple droplets. Some of the nozzles that we used to generate multiple droplets are shown in figure 3(a), and the corresponding size distribution is shown in figure 3(b). As shown in the figure, the generated droplets show a polydisperse size distribution, with a radius varying from 50 to 250 microns. We used the high-speed shadowgraphy technique at 4000 Hz to characterize the droplets. The shadowgraphy image showing the typical droplet distribution, from one of the tested nozzles (6-hole nozzle), is shown in figure 3(c).

III. RESULTS AND DISCUSSION

The primary focus of the present study is the interaction of droplets with vortical structures to gain a better understanding of droplet dynamics in atmospheric clouds. As discussed in the previous section, we generate multiple droplets and a single vortical structure to study their inter-



(a) (b) (c)

Figure 3: **(a) Various nozzles tried to generate multiple droplets; (b) size distribution of different nozzles used to generate multiple droplets; (c) shadowgraphy image showing the typical droplet distribution from a 6-hole mist-type nozzle.**

action. In real clouds, however, multiple vortices exist; but due to the challenges involved in generating such conditions in a laboratory setting, the present study considers a single idealized vortex, which follows a Gaussian vorticity distribution along the core [6]. Further, in the atmospheric clouds, the droplet Collision-Coalescence is typically influenced by the dissipation-range ($6\eta - 60\eta$, $\eta = 1mm$ for clouds) scales of air turbulence. Therefore, in the present study, the droplet size distribution, vortex size (2.7 cm), and vortex strength (5080 cm^2/s) were chosen such that the resulting Stokes number falls within a range similar to that in atmospheric clouds. We then examine how the vortex affects droplet trajectories, how it leads to accumulation in low-vorticity regions, and ultimately how it promotes collision and coalescence in a turbulent environment.

The shadowgraphy technique was employed to study the interaction of multiple droplets with an idealised vortical structure. After the interaction, one can expect a preferential concentration of droplets where droplets will migrate from high to low vorticity regions, leading to the generation of caustics and clustering [7]. We utilize the Voronoï analysis to identify these clusters along with the voids generated after the interaction [8], [9]. A typical voronoï diagram for the droplets after the interaction is shown in figure 4 (a), where the blue cells represent the clusters and the green cells represent the voids, generated due to the influence of the vortex. We classified clusters and voids with reference to the Random Poisson Process (RPP) distribution, where droplets are assumed to be uniformly distributed. To enable comparison, the PDFs of the RPP distribution and the experimental normalized voronoï areas are plotted together in figure 4(b). For the experimental data, we calculated the voronoï cell areas in each frame and normalized them by the frame mean, then combined all valid samples to compute $x = A/\langle A \rangle$. The experimental PDF was then obtained using fixed log-spaced bins, while uncertainty was assessed with a nonparametric bootstrap approach. In this procedure, 1000 bootstrap samples of size n were created by resampling x with replacement. For each sample, the PDF was recalculated, and the pointwise 95% confidence bands were obtained from the 2.5–97.5th percentile range of the resampled distributions in each bin. The resulting figure therefore shows the PDF of the pooled normalized voronoï areas ($x = A/\langle A \rangle$) over the entire time interval, with the 95% bootstrap confidence interval indicated by the gray band. For reference, the random distribution of droplets predicted by the Random Poisson Process (RPP) is also plotted. Two vertical lines at intersections A_1 and A_2, is the locations where Experimental PDF crosses the RPP pdf, defining three different regimes: Clusters ($A < A_1$), Regulars ($A_1 \leq x \leq A_2$), and Voids ($x > A_2$). The voronoï area is inversely proportional to the number density of the droplets. Therefore, if the normalized area is less than the first intersection point, it can be classified as a cluster, and voronoï areas greater than the second intersection point can be classified as voids. A clear distinction between clusters and voids is evident in figure 4(b), indicating that the vortex drives droplets to cluster in regions of low vorticity. To quantify this clustering, we compare the standard deviation of the experimental voronoï PDF with the RPP PDF and compute the degree of clustering, $\sigma_{\mathrm{rel}} = (\sigma_v - \sigma_{\mathrm{RPP}})/\sigma_{\mathrm{RPP}}$, over the time domain as the vortex travels from the left to the right of the field of view. It can be seen from figure 4(c) that the degree of clustering increases as the vortex propagates. For the present case, we find a mean degree of clustering of $\sigma_{\mathrm{rel}} = 0.74$, which is larger than $\sigma_{\mathrm{RPP}} = 0.53$ corresponding to randomly distributed droplets [10]. Therefore, the vortex leads to clustering of droplets in low-vorticity regions, and this process may lead to the formation of caustics, where droplets can have multi-valued velocities at a given location, thereby enhancing collision rates. Eventually, this may lead to the caustics-induced coalescence in the atmospheric clouds, which promotes droplet growth [11].

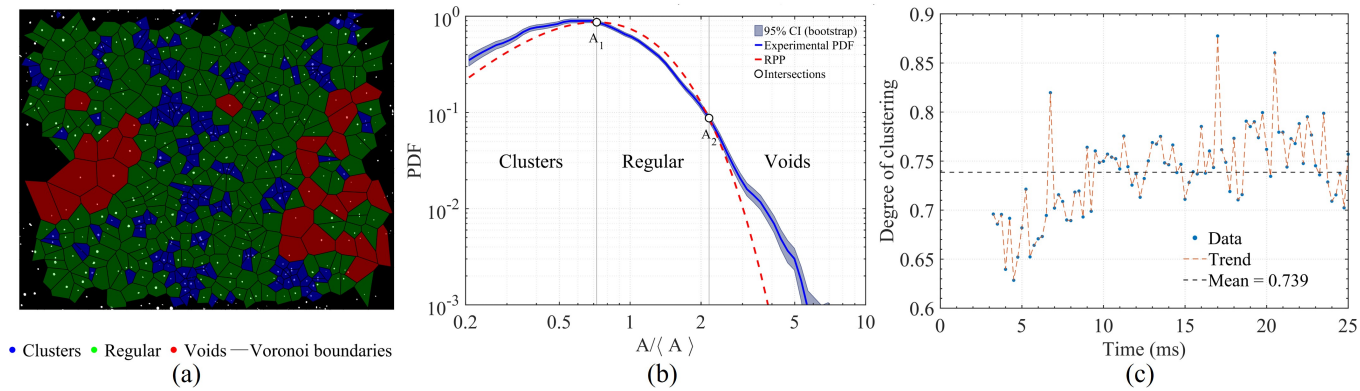


Figure 4: **(a) voronoï diagram showing the clusters and voids generated after the vortex influences the droplets; (b) Comparison of PDFs of experimental normalized voronoï area with respect to the random Poisson process (RPP) distribution; (c) Variation of degree of clustering,** $\sigma_{\mathrm{rel}} = (\sigma_v - \sigma_{\mathrm{RPP}})/\sigma_{\mathrm{RPP}}$**, over the time domain as the vortex travels from the left to the right of the field of view.**

IV. CONCLUSIONS

We experimentally investigated the interaction of droplets with a vortical structure to better understand the growth of cloud droplets in atmospheric clouds. Polydisperse droplets with radii ranging from 50 to 250 microns were generated, and their interaction with a single vortical structure was studied to examine the influence of background turbulence on droplet dynamics. We employed the high-speed shadowgraphy technique to study the interaction and observed that following the interaction, a vortex gives rise to clusters and voids, which may lead to the formation of caustics. These caustics may further enhance the collision rates and subsequently promote the coalescence of cloud droplets in atmospheric clouds.

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